

Chapter 8.A

The Quantum Model of the Atom

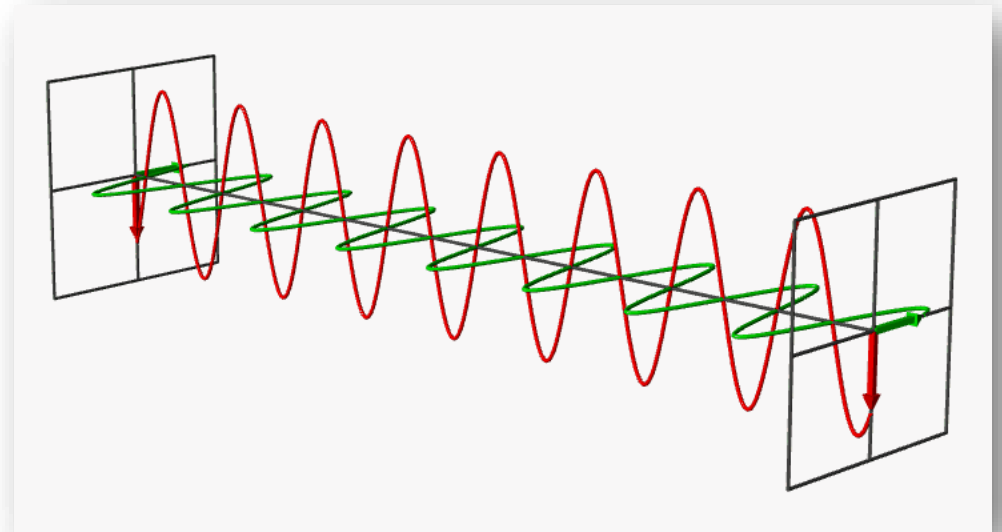
Chemical Principles I

CHEM 1211K

1. Describe and apply the dual nature of light.
2. Calculate energy, frequency, or wavelength of a sample of light.
3. Use the Bohr model of energy levels in atoms to explain light emission and absorption by gaseous atoms.
4. Calculate the energy or wavelength of a specific electronic transition in a hydrogen atom.
5. Use the concept of quantum numbers to characterize the energies, spatial distributions, and spins of atomic electrons.
6. Visualize and describe the spatial distributions and energies of atomic electrons using the quantum model of the atom.
7. Sketch the electronic energy levels of an atom.
8. Add electrons to atomic orbital energy (energy level) diagrams.
9. Determine electron configurations for the elements using the concepts of shells, subshells, and orbitals.

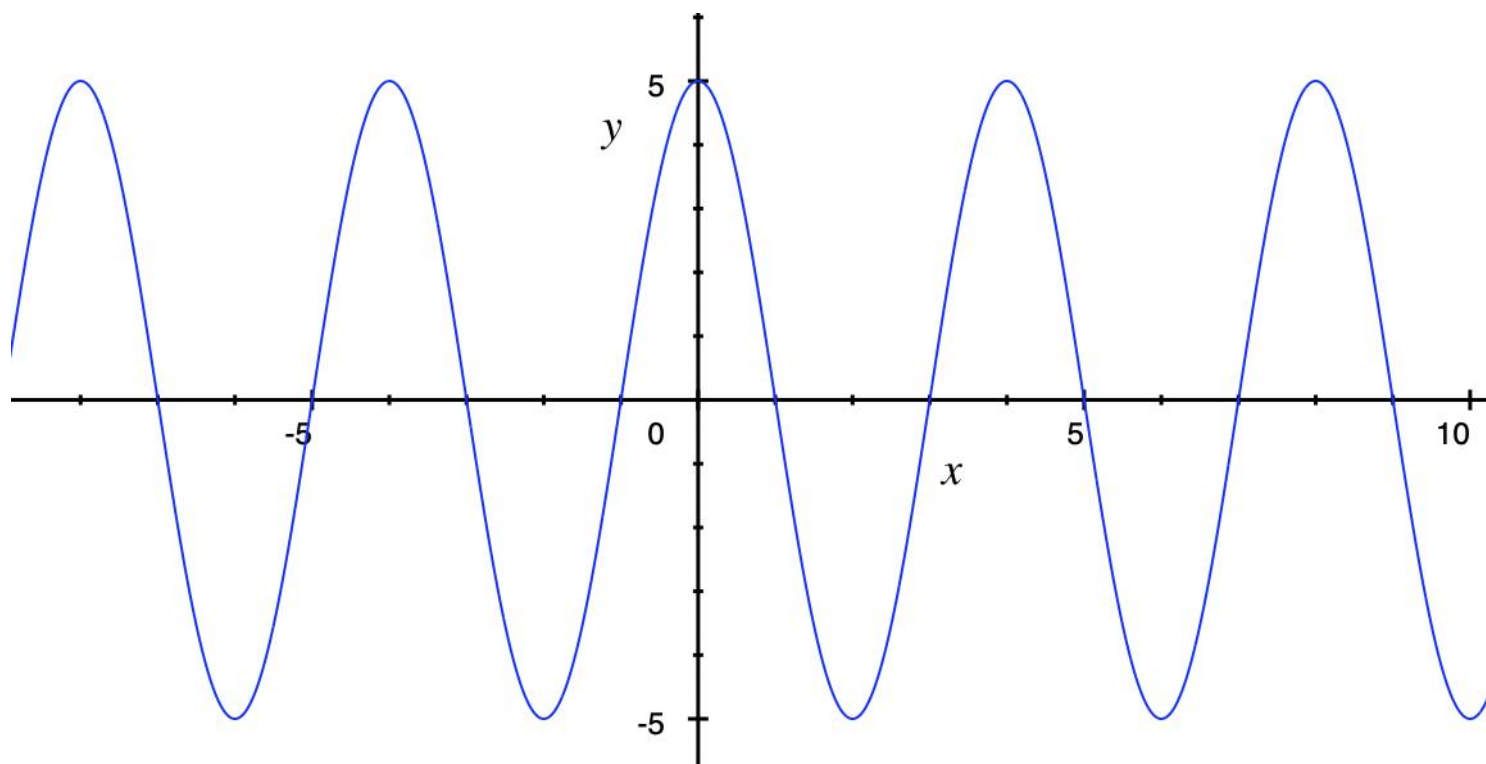
8.1 Classical Model of Light

- **Light** is classically viewed as an oscillating electromagnetic wave. The lengths and energies of light waves vary *enormously*.
- The electric field (red) interacts with charged particles, particularly electrons in atoms and molecules.
- A magnetic field (green) oscillates at right angles to the electric field.



8.1 Properties of Light Waves

- Waves have some important spatial characteristics with relevance to chemistry...
 - **Wavelength (λ):** the distance between corresponding points on adjacent waves (units of length)
 - **Frequency (ν):** the number of oscillations of the wave per unit time (units of time^{-1})
 - **Amplitude:** the maximum “height” of a wave above the zero line (units of length)
 - **Speed:** the rate at which a fixed point on the wave travels through space (units of $\text{length} \cdot \text{time}^{-1}$)



8.1 Frequency, Wavelength, and Speed



- The **speed of light (c)** has a constant value of 2.998×10^8 m/s.
- Speed, frequency, and wavelength are theoretically related. Because the speed of light is a constant, wavelength and frequency are inversely related.

$$c = \lambda \times \nu$$

$$\nu = \frac{c}{\lambda}$$

- These equations will become very important when we relate light frequency to another important property.

8.1 The Electromagnetic Spectrum

- Matter can absorb light; classically this was viewed as a decrease in amplitude as a light wave passes through a material.

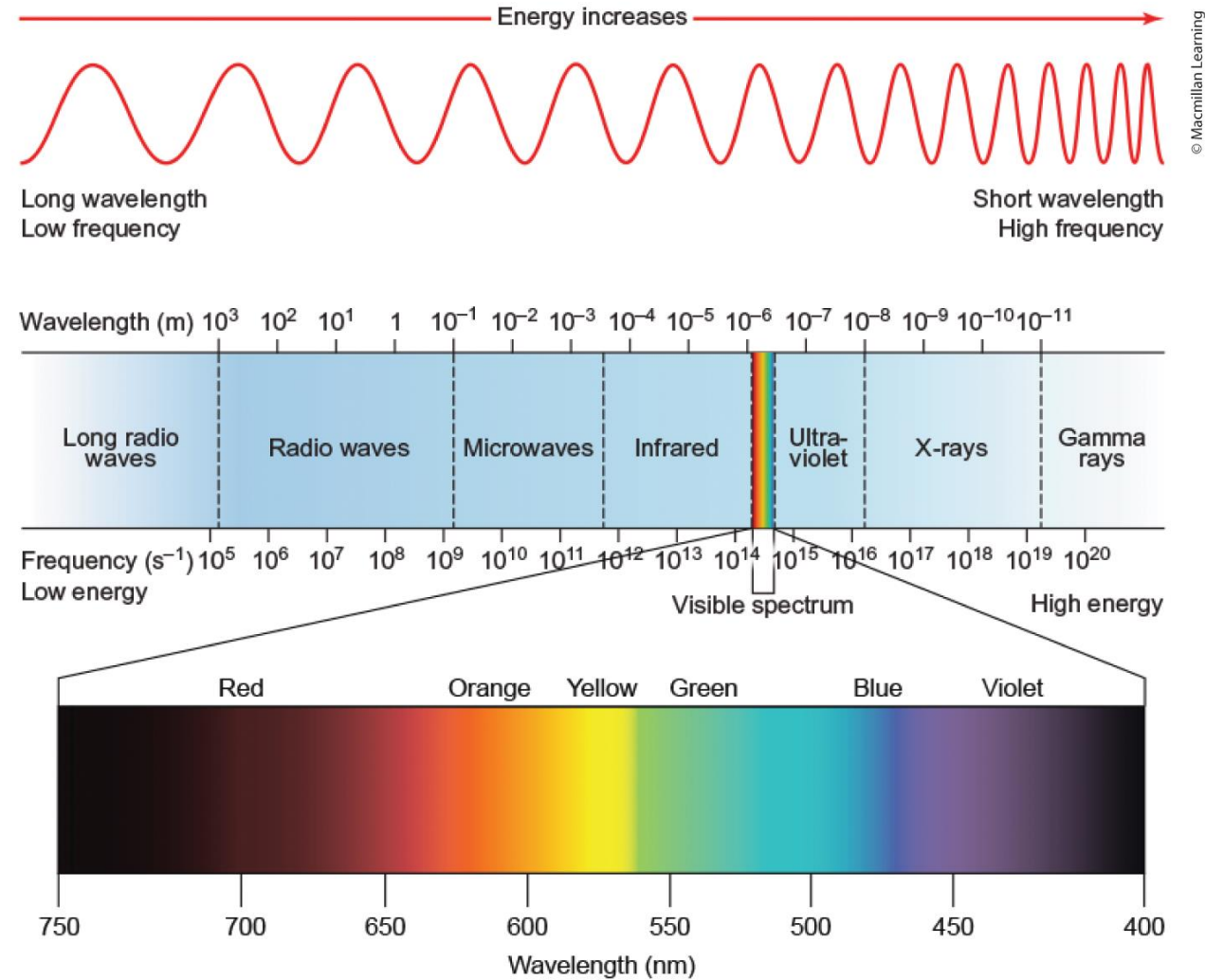
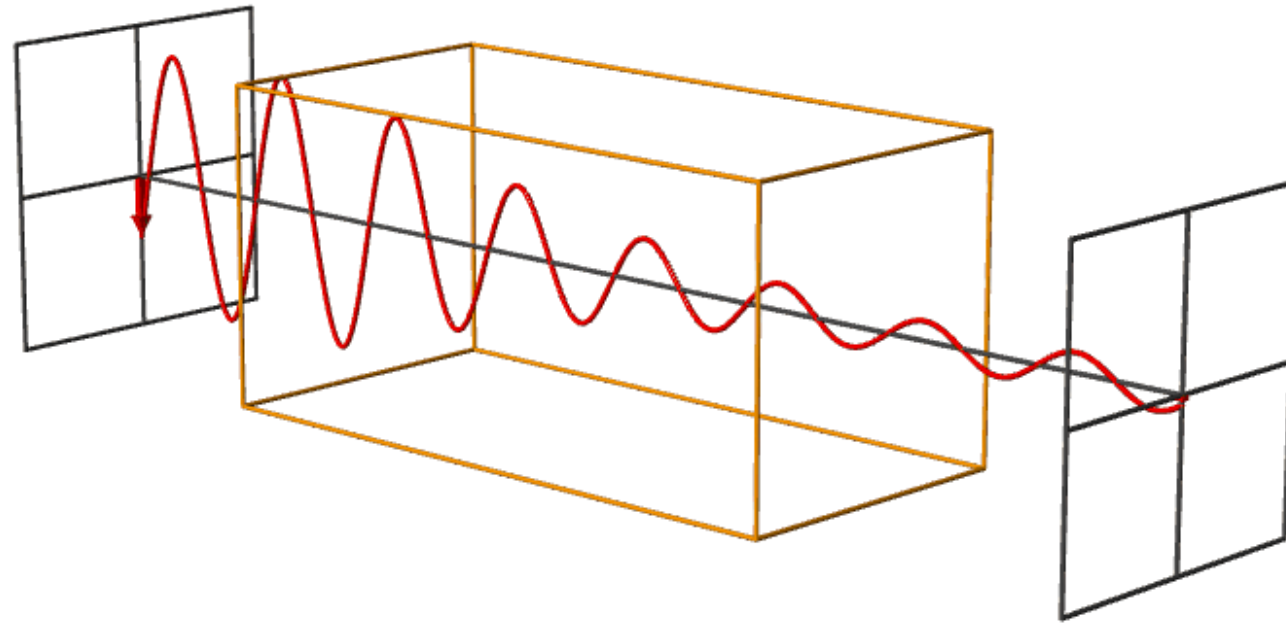


Figure 8.4. The electromagnetic spectrum.

8.1 The Interaction of Light with Matter

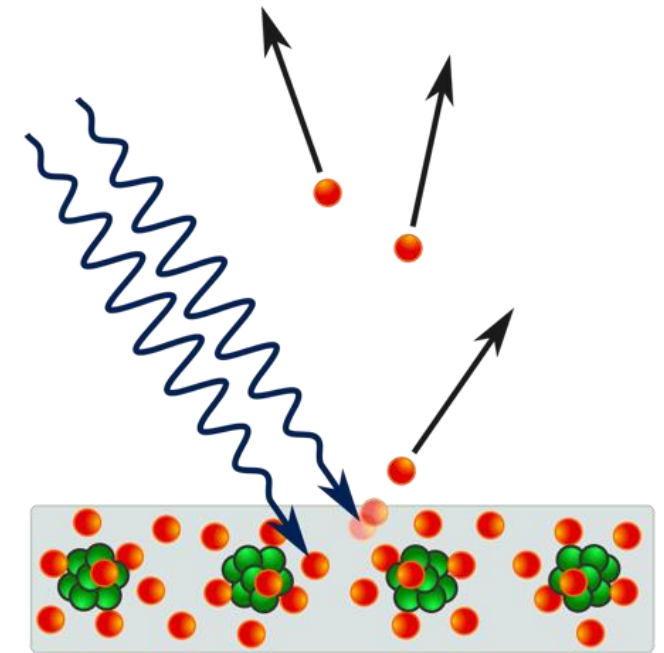
- Matter can **absorb** light; classically this was viewed as a decrease in amplitude as a light wave passes through a material.



$$A = \log_{10} \frac{I_0}{I}$$

8.1 The Photoelectric Effect

- The **photoelectric effect** was not well described by the wave model of light; from it, a particle model of light was developed by Einstein and others.
 - Light impinging on a metal causes ejection of electrons from the surface and a measurable photocurrent and/or voltage.
 - Ejected electrons are only observed when the light is *above* a threshold frequency ν_{min} .
 - The kinetic energy of each ejected electron is unaffected by light intensity but scales linearly with frequency (above ν_{min}).
 - Light intensity affects only the number of ejected electrons per unit time.



8.1 Photons; the Particulate Model of Light



- A description of light as a collection of particles (**particulate model**) explains the photoelectric effect well.
- The particulate model of light postulates that it is composed of tiny massless particles called **photons**.
 - The energy of a photon is directly proportional to its frequency, with **Planck's constant** ($h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$) the constant of proportionality.
 - Intense or bright light contains more photons than dim light.
- For a collection of N photons, each with frequency ν and wavelength λ ,

$$E = Nh\nu = N\frac{hc}{\lambda}$$

The electron-Volt (eV) is a convenient unit for the energies of single photons. The amount of energy required to “push” an electron through a potential difference of 1 Volt “uphill” corresponds to 1 eV. Planck’s constant is equal to $4.14 \times 10^{-15} \text{ eV}\cdot\text{s}$.

Calculate the wavelength of a single photon with an energy of 1.00 eV. Which of the following best describes the scale of this wavelength?

- A. Picometers
- B. Nanometers
- C. Micrometers
- D. Millimeters
- E. Meters

The amount of energy we calculated in the previous problem is extremely tiny. Convert this energy to Joules, recalling that Planck's constant is 6.626×10^{-34} J·s. What is the energy of one mole of photons with energy 1.00 eV in kilojoules per mole (kJ/mol)? Enter your response to the nearest 0.1 kJ/mol.

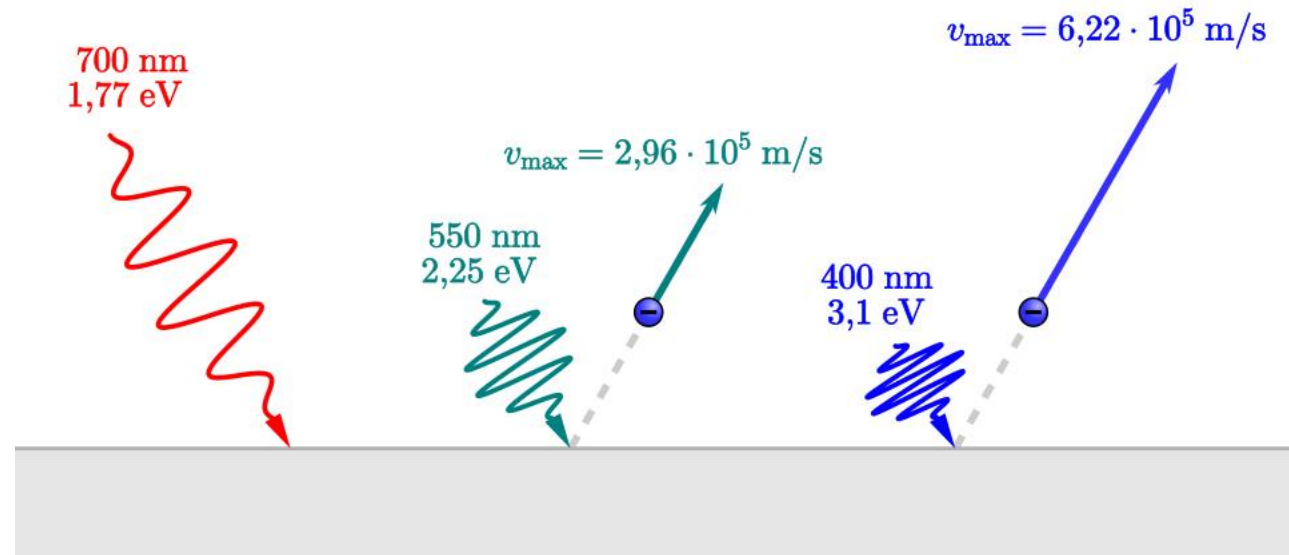
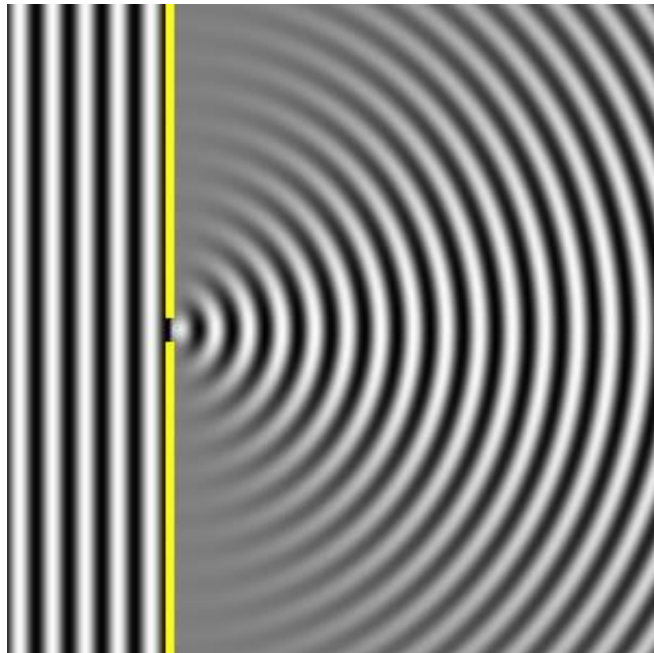
A laser pulse with wavelength 532 nm contains 0.00467 J of energy.
How many photons are in the laser pulse?

Planck's constant is 6.626×10^{-34} J·s and the speed of light is 2.998×10^8 m·s⁻¹.

- A. 1.50×10^{24}
- B. 1.25×10^{16}
- C. 3.74×10^{-19}
- D. 1.25×10^{25}
- E. 3.74×10^{-28}

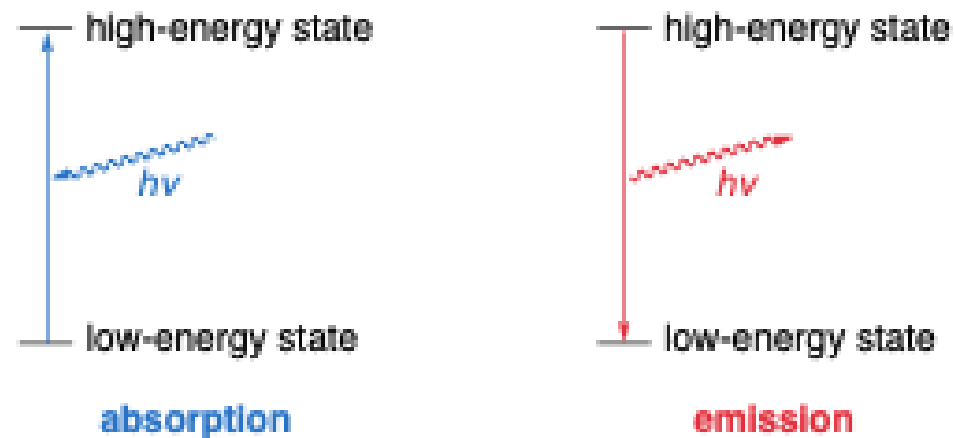
8.1 Wave-particle Duality

- How can a particle have a wavelength? Well, light is not purely a wave nor purely a particle...the nature of light incorporates aspects of both. Light displays **wave-particle duality**.
 - In some experimental circumstances, light acts like a wave: diffraction, refraction, and interference, for example.
 - In others, light acts like a particle: photoelectric effect and absorption and emission by matter, for example.



8.2 Interaction of Light and Matter

- Atoms and molecules can absorb and emit light, with changes in energy equal to the energy of the photon absorbed or emitted.
- The particulate model of light is best for our understanding here.
- **Light absorption** occurs when an atom or molecule takes in a photon; the atom or molecule increases in energy by an amount $E = h\nu$.
- **Light emission** occurs when an atom or molecule gives off a photon; the atom or molecule decreases in energy by an amount $E = h\nu$.
- Absorption and emission are commonly represented on **energy-level diagrams**.



8.2 Atomic Absorption and Emission Spectra

- Each element has a unique set of wavelengths of light that atoms of the element absorb; the intensity of absorption as a function of wavelength of the light is called the **absorption spectrum**.
- The same wavelengths are involved when “excited” atoms of an element emit photons; the intensity of emission as a function of wavelength of the light is called the **emission spectrum**.

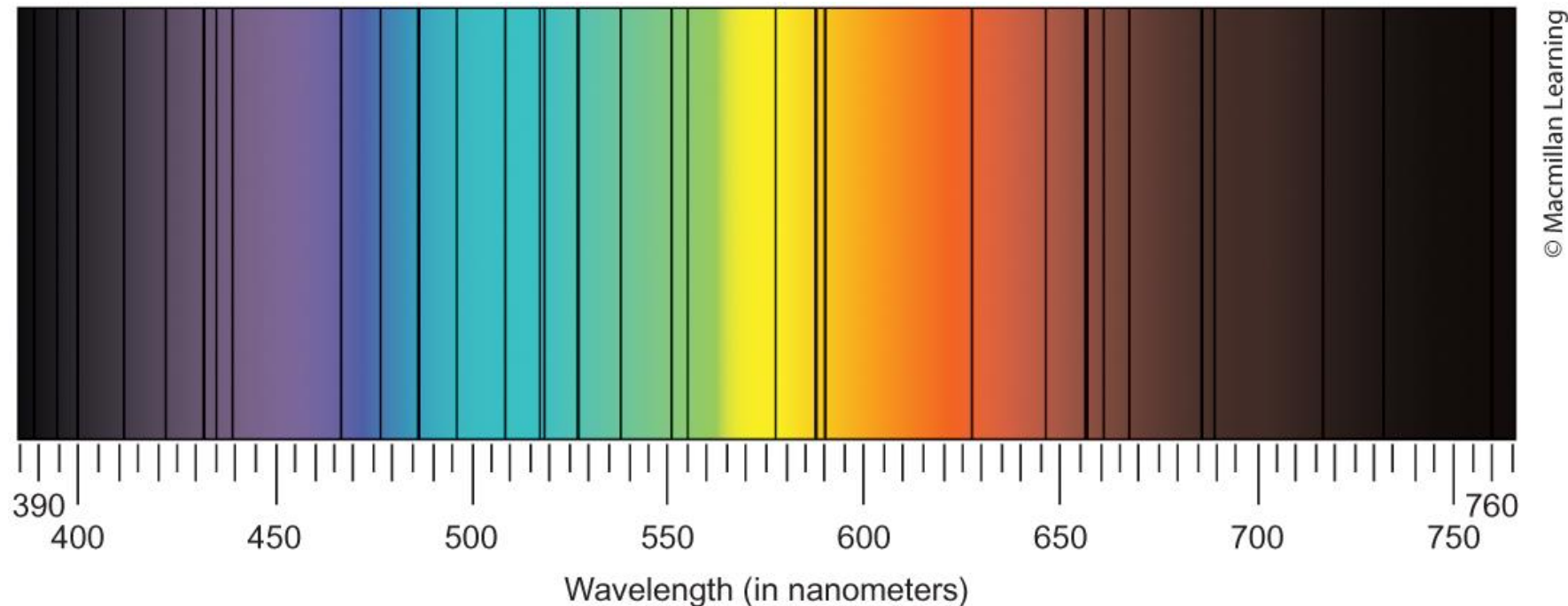


Figure 8.8. The solar spectrum. Missing wavelengths are due to absorption by solar elements, particularly hydrogen and helium.

- What does this suggest about the nature of energy in atoms?! We'll discuss the implications next class.

Chapter 8.B

The Quantum Model of the Atom

Chemical Principles I

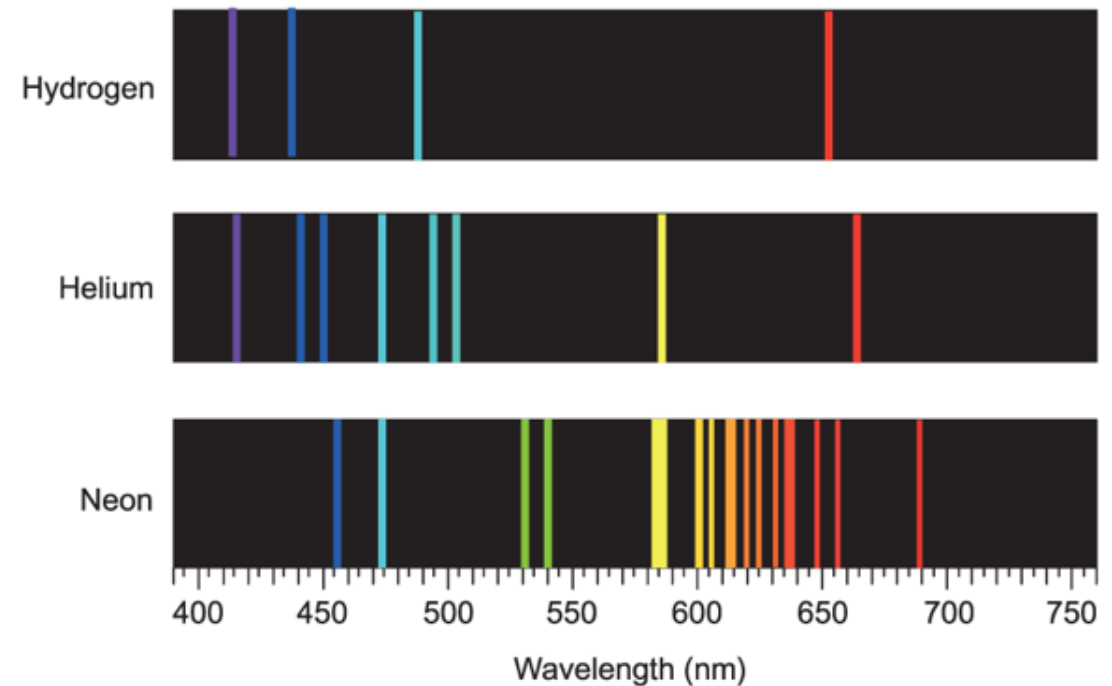
CHEM 1211K

1. Describe and apply the dual nature of light.
2. Calculate energy, frequency, or wavelength of a sample of light.
3. Use the Bohr model of energy levels in atoms to explain light emission and absorption by gaseous atoms.
4. Calculate the energy or wavelength of a specific electronic transition in a hydrogen atom.
5. Use the concept of quantum numbers to characterize the energies, spatial distributions, and spins of atomic electrons.
6. Visualize and describe the spatial distributions and energies of atomic electrons using the quantum model of the atom.
7. Sketch the electronic energy levels of an atom.
8. Add electrons to atomic orbital energy (energy level) diagrams.
9. Determine electron configurations for the elements using the concepts of shells, subshells, and orbitals.

8.2 Atomic Spectroscopy

- Gaseous atoms of an element heated or excited with electricity emit very specific wavelengths of light; the intensity of emission as a function of wavelength is called the **atomic emission spectrum**.
- **Atomic spectroscopy** is an analytical technique used to measure the absorption or emission spectra of the elements. [This interactive](#) can be used to play around with this technique.

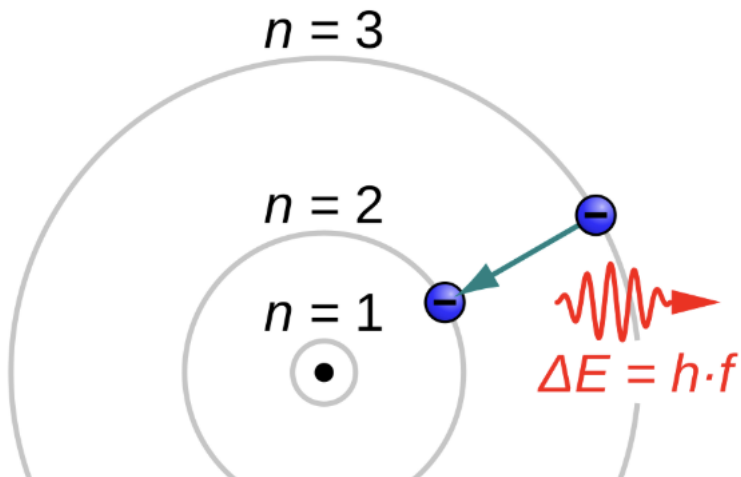
- Why do atoms emit only **specific wavelengths** of light?
- Why does each element have a **unique** emission spectrum?
- **What's happening inside the atom** during emission?



8.2 The Bohr Model of the Atom

- The **Bohr model** of the atom doesn't exactly answer these questions but does provide a description of the atom that accounts for discrete (non-continuous) spectra.
 - Atomic electrons can only “occupy” certain energy levels with finite spacing between them.
 - Electrons travel in circular orbits around the nucleus. The higher the energy level, the greater the radius of orbit.
 - When a photon is absorbed or emitted, an electron transitions from one energy level to another. The energy of the photon absorbed or emitted matches the gap between energy levels exactly.

$$\Delta E = E_2 - E_1 = h\nu$$



8.2 Ground and Excited States

- In the lowest energy level (**ground state**), the electron is closest to the nucleus.
- Absorption of a photon “promotes” the electron to a higher energy level with a greater electron-nucleus distance; the atom is said to be in an **excited state**.
- Excited-state atoms can emit a photon to return to the ground state. These emitted photons appear in the emission spectrum.

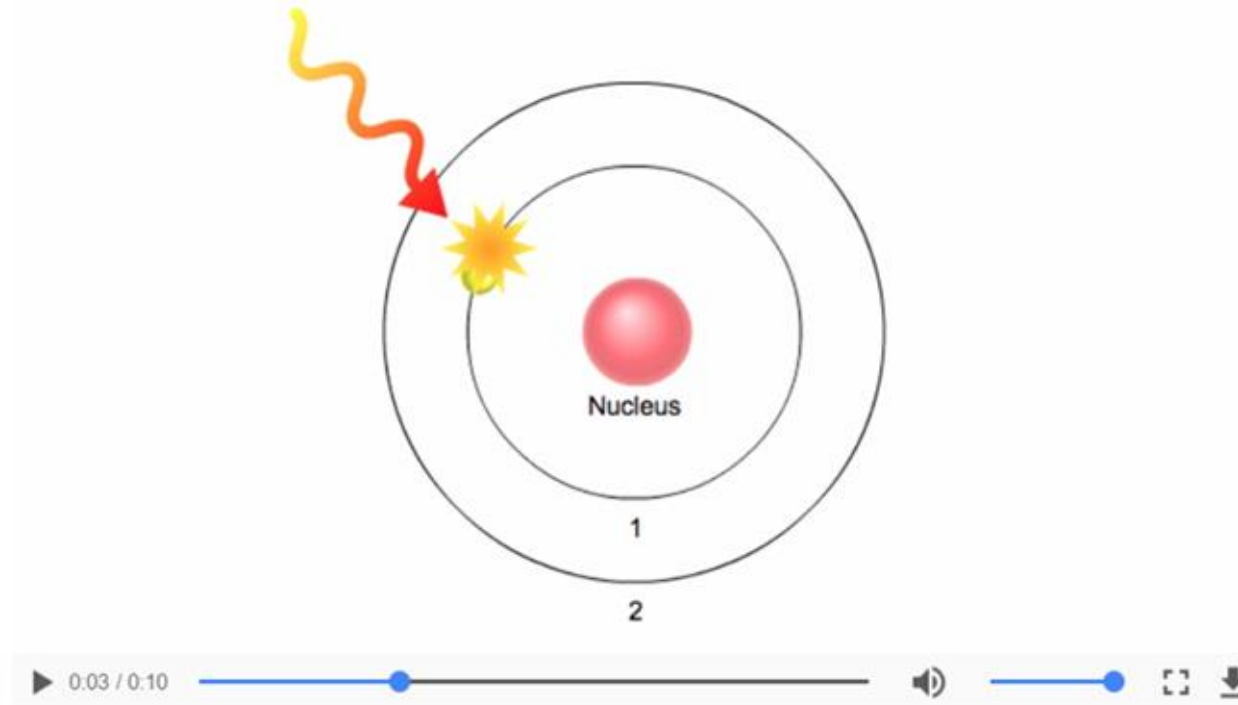


Figure 8.10. The Bohr model of the atom and light absorption and emission.

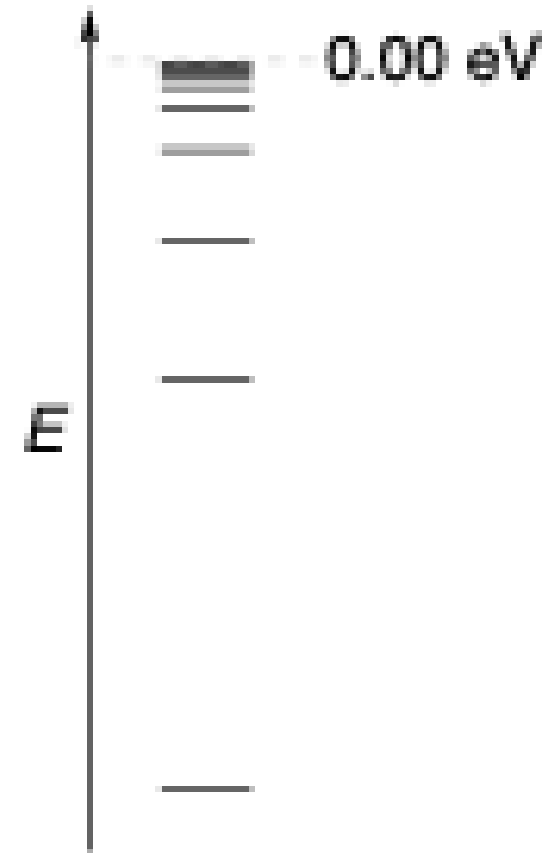
8.2 Absorption/Emission Wavelengths of the Hydrogen Atom

- Most wavelengths of light emitted by atoms do not follow simple mathematical patterns. However, the wavelengths emitted by hydrogen atoms do.
- The **Rydberg equation** describes the energies and wavelengths of photons emitted or absorbed by hydrogen. For a transition between two energy levels n_1 and n_2 ,

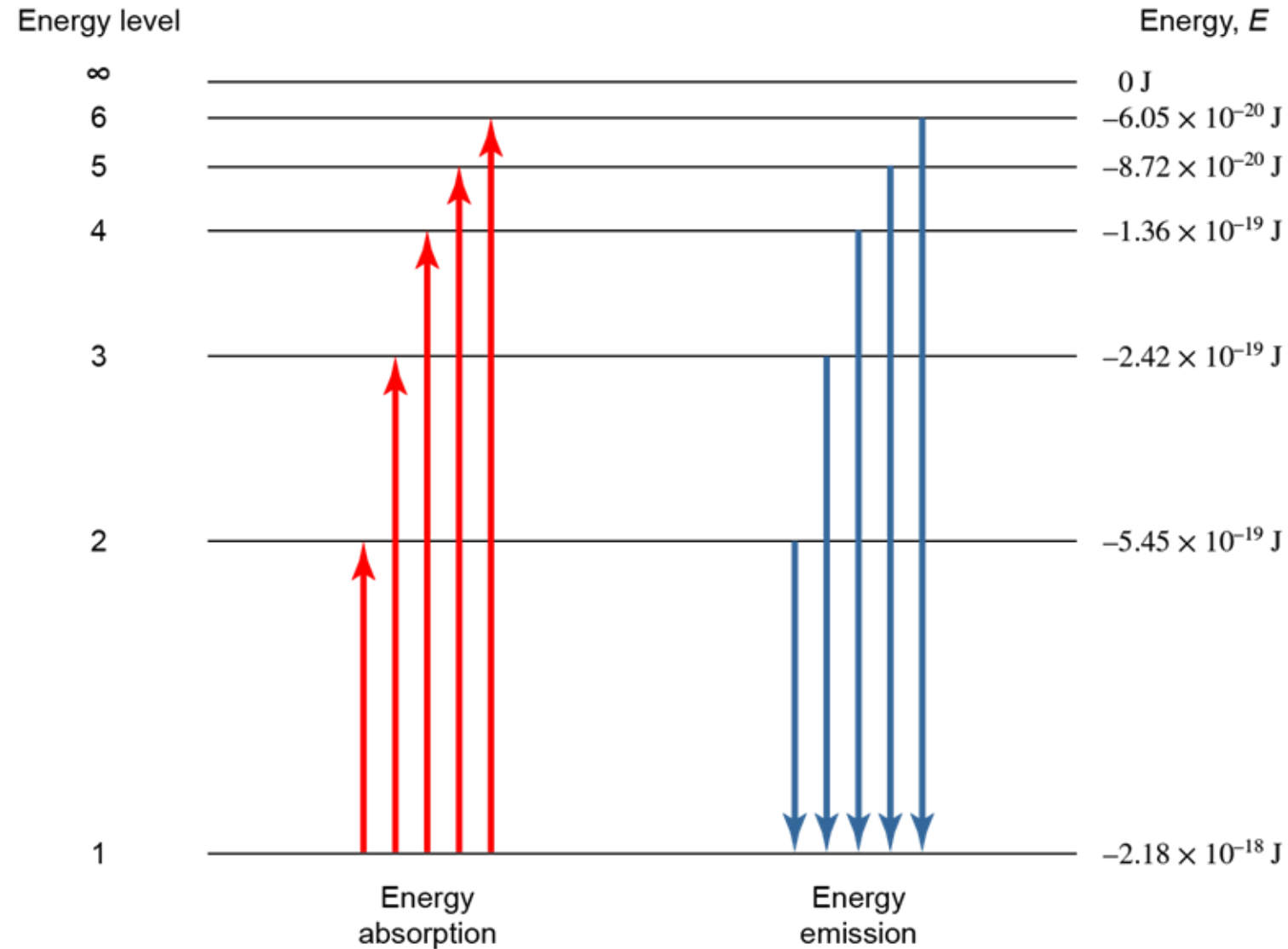
$$E(n) = -R_{\infty} \left(\frac{1}{n^2} \right)$$

$$\Delta E = E(n_2) - E(n_1) = -R_{\infty} \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

$$R_{\infty} = 13.6 \text{ eV} = 2.18 \times 10^{-18} \text{ J}$$



8.2 Electronic Transitions on an Energy Diagram



Try it yourself!

Figure 8.13. Electronic energy levels and transitions between energy levels in the hydrogen atom.

8.2 Wavelength-based Version of the Rydberg Equation

- An equivalent formulation of the Rydberg equation uses reciprocal wavelength instead of energy. The two are proportional with constant of proportionality hc !

$$\frac{1}{\lambda} = R'_{\infty} \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

$$R'_{\infty} = \frac{R_{\infty}}{hc} = 1.097 \times 10^7 \text{ m}^{-1}$$

Example. Determine the wavelength (in nanometers, nm) of the photon emitted by a hydrogen atom when its electron jumps from $n = 3$ to $n = 2$.

8.2 Notes on Electronic Energy Levels



- Some notes on electronic energy levels in the Bohr model (for all elements)...
 - The largest energy gap is between the ground state and the first excited state.
 - Energy levels get more closely spaced with increasing n and distance from the nucleus.
 - Complete dissociation of the electron from the atom (**ionization**) is associated with $E = 0$.
 - **Bound states** of the electron have negative energy; an input of energy is required for ionization.

Which of the following electronic transitions for the hydrogen atom will produce the spectral line with the shortest wavelength?

- A. $n = 5$ to $n = 4$
- B. $n = 2$ to $n = 1$
- C. $n = 1$ to $n = 0$
- D. $n = 3$ to $n = 2$
- E. $n = 4$ to $n = 3$

In the emission spectrum of sodium, the most prominent line is a yellow one at 589 nm. What is the energy of this photon in electron-Volts (eV)? The product of Planck's constant and the speed of light (hc) is 1240 eV·nm.

The energy of a photon emitted by an excited hydrogen atom was measured as 1.9 eV. Which of the following statements is/are necessarily *true*? Select all that apply.

- A. This measurement indicates that one of the energy levels of the hydrogen atom is at -1.9 eV.
- B. This measurement indicates that there is a gap between two energy levels of 1.9 eV.
- C. After emission, the atom is in a lower-energy state than before emission.
- D. After emission, the electron is farther from the nucleus than before emission.
- E. After emission, the atom is in the ground state.

Chapter 8.C

The Quantum Model of the Atom

Chemical Principles I

CHEM 1211K

1. Describe and apply the dual nature of light.
2. Calculate energy, frequency, or wavelength of a sample of light.
3. Use the Bohr model of energy levels in atoms to explain light emission and absorption by gaseous atoms.
4. Calculate the energy or wavelength of a specific electronic transition in a hydrogen atom.
5. Use the concept of quantum numbers to characterize the energies, spatial distributions, and spins of atomic electrons.
6. Visualize and describe the spatial distributions and energies of atomic electrons using the quantum model of the atom.
7. Sketch the electronic energy levels of an atom.
8. Add electrons to atomic orbital energy (energy level) diagrams.
9. Determine electron configurations for the elements using the concepts of shells, subshells, and orbitals.

8.3 Wave-Particle Duality Generalizes to Matter

- We've seen that light has wave properties and particle properties; *matter* also exhibits wave-particle duality at very small scales!
- Electrons in atoms display wave-like and particle-like behavior, depending on the situation...
 - Certainty in position (x) leads to a particulate description of the electron.
 - Certainty in momentum ($p = mv$) leads to a wave description of the electron.
- The **Heisenberg uncertainty principle** expresses fundamental limits on the precision of position and momentum measurements.

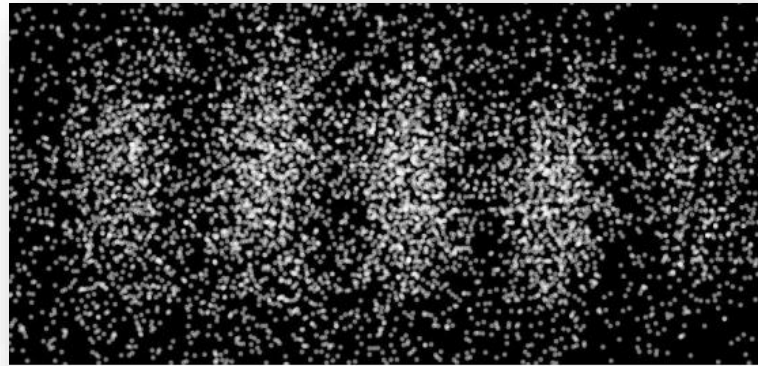
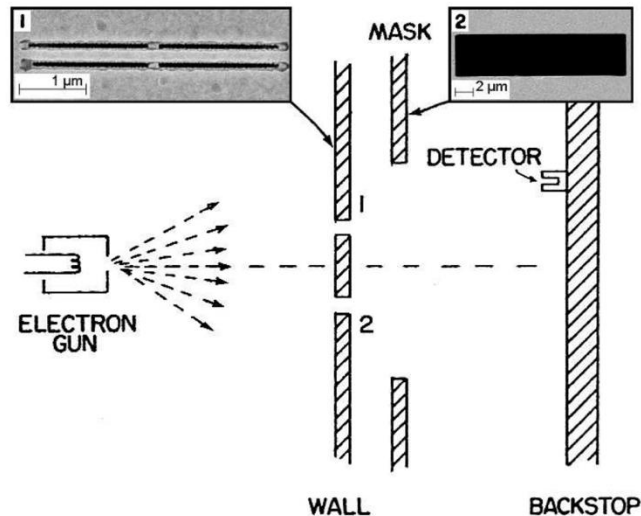


Figure. After passing through two closely spaced slits, electrons display a wave-like diffraction pattern. <https://arxiv.org/pdf/1210.6243>

- Each electron “carries” wave-like information with it. Equivalently, there is a wave-like description of the electron embedded within the larger physical system.
- The **Schrödinger equation** is based on a wave-like description of physical systems; solving it for the hydrogen atom gives a wave-like description of the electron (the “wave-like information” that the electron “carries”).

$$\hat{H}\Psi = E\Psi$$

- H is the **Hamiltonian operator**, which includes terms for kinetic and potential energy of the electron(s). H is specified by the physical system.
- Ψ is the **wave function**, a complex-valued function over all space, which “encodes” all measurable information about the electron(s). Most notably, it provides the probability $|\Psi|^2$ of finding an electron at each point in space.
- E is the total **energy**, a measurable value.
- Solutions $\{\Psi_i, E_i\}$ include a wave function and paired energy value. There are infinitely many solutions, but the E_i values are finitely spaced and there is a minimum E_1 !

8.4 Atomic Orbitals are Solutions to the Schrödinger Equation



- Electrons in atoms are not particles confined to circular orbits as in the Bohr model; their dynamics are defined by probabilistic **atomic orbitals Ψ** .
- Every orbital has three important properties...
 - **Occupancy.** How many electrons are described by (“are in”) this orbital?
 - **Energy.** How stable are electrons in this orbital?
 - **Shape.** What is the spatial probability distribution of electrons in this orbital?

8.4 Quantum Numbers “Describe” the Atomic Orbitals

- For the hydrogen atom, atomic orbitals are organized into various “layers” associated with parameters in the mathematical expressions for the atomic orbitals. These parameters, which act like “names” for the atomic orbitals, are known as **quantum numbers**.
- **Shells** are associated with the quantum number n . We have previously called shells **energy levels**.
- Most shells contain *multiple* atomic orbitals; within a shell, orbitals are organized into **subshells** by quantum number ℓ .
- Each subshell may contain multiple orbitals, each of which has a unique quantum number m_ℓ .

ℓ	Subshell	Number of Orbitals	Maximum Number of e^- in Subshell
0	<i>s</i>	1	2
1	<i>p</i>	3	6
2	<i>d</i>	5	10
3	<i>f</i>	7	14

Table 8.2. Names and characteristics of the atomic subshells.

- The four atomic quantum numbers act like an “address” for electrons in the atom.
- The solutions to the Schrödinger equation for the hydrogen atom...

$$\psi(n, \ell, m_{\ell}, m_s)$$

- The **Pauli exclusion principle** states that, within an atom, no two electrons can share the same set of quantum numbers. Each electron has a unique “address.”

8.4 The Principal Quantum Number n



- **Principal quantum number (n)**
- May have any positive integer value (1, 2, 3, ...)
- Larger values of n ...

8.4 The Orbital Quantum Number ℓ

- **Orbital (angular momentum) quantum number (ℓ)**

ℓ	Subshell
0	<i>s</i>
1	<i>p</i>
2	<i>d</i>
3	<i>f</i>

- Allowed values of ℓ depend on n ...
- The quantum numbers n and ℓ are often combined to create orbital or subshell labels.

Consider the constraint on ℓ . How many subshells are present in the shell with $n = 3$?

Enter your response as a numeral.

8.4 The Magnetic Quantum Number m_ℓ



- **Magnetic quantum number (m_ℓ)**
- May have any integer value from $-\ell$ to $+\ell$; each possible value corresponds to an orbital in the subshell.
- **Example.** What values of m_ℓ are possible for an electron with $n = 3$ and $\ell = 2$?

8.4 The Spin Quantum Number m_s



- The **spin quantum number (m_s)** indicates the “direction” of the magnetic moment of the electron created by its intrinsic angular momentum. There are only two possible values: $+1/2$ and $-1/2$.
- Electrons don't really [spin](#); angular momentum is just an intrinsic property of the electron.

From the choices below, select the two subshells that contain the same number of orbitals (or, if none do, select choice E).

- A. $2s$
- B. $3d$
- C. $4d$
- D. $2p$
- E. Each of these subshells contains a different number of orbitals.

How many orbitals in an atom may have the combination of quantum numbers $n = 5$ and $m_l = -3$?
Consider all possible or allowed subshells!

Chapter 8.D

The Quantum Model of the Atom

Chemical Principles I

CHEM 1211K

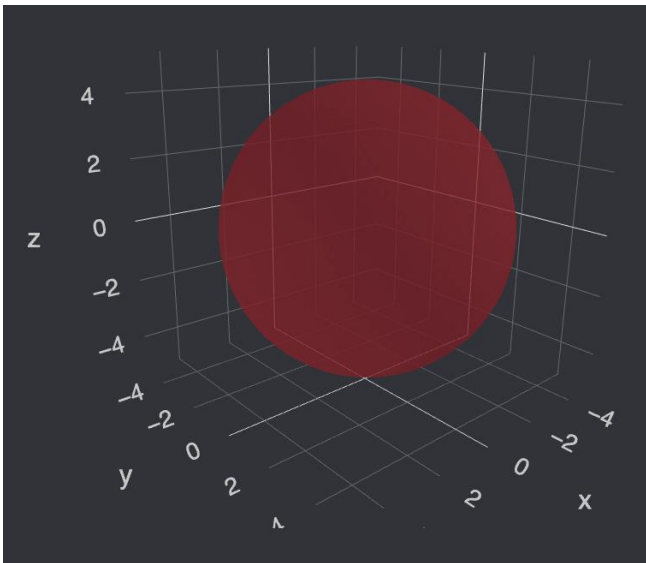
1. Describe and apply the dual nature of light.
2. Calculate energy, frequency, or wavelength of a sample of light.
3. Use the Bohr model of energy levels in atoms to explain light emission and absorption by gaseous atoms.
4. Calculate the energy or wavelength of a specific electronic transition in a hydrogen atom.
5. Use the concept of quantum numbers to characterize the energies, spatial distributions, and spins of atomic electrons.
6. Visualize and describe the spatial distributions and energies of atomic electrons using the quantum model of the atom.
7. Sketch the electronic energy levels of an atom.
8. Add electrons to atomic orbital energy (energy level) diagrams.
9. Determine electron configurations for the elements using the concepts of shells, subshells, and orbitals.

8.5 Representing the Shapes of Orbitals

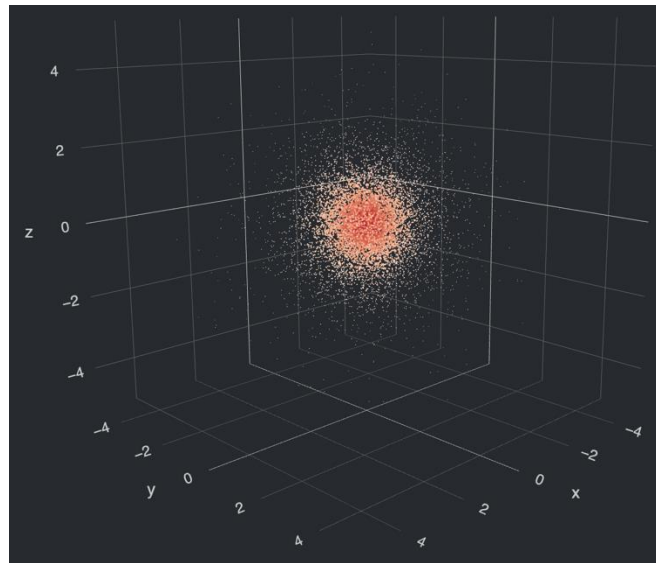
- Orbitals are functions over three-dimensional space representing the probability of finding an electron at each point in space.
- How do we represent a function of three variables using only three spatial dimensions?
 - Isosurface:** a surface at which the value of ψ is constant, and inside of which the probability of finding an electron is high.
 - Sampled representation:** we “look for an electron” in the orbital many times and plot the result.
 - Radial distribution:** plots the value of ψ along a radial line starting at the nucleus.



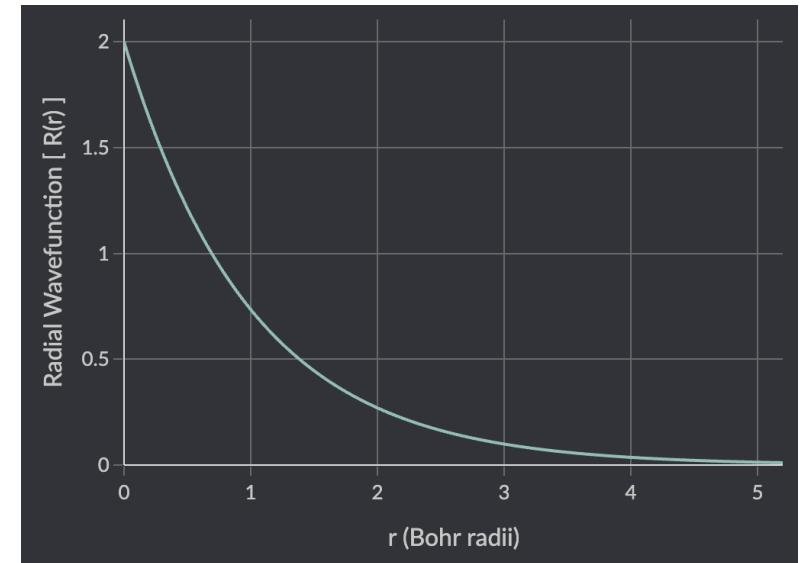
Try it yourself!



Isosurface



Sampled



Radial Distribution

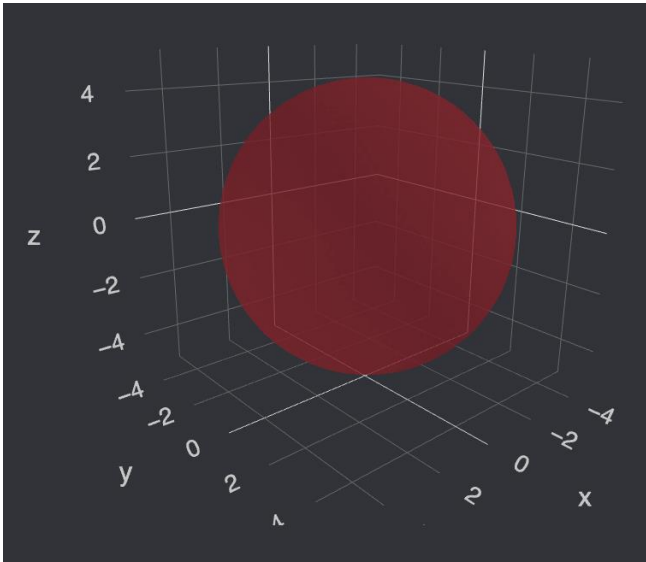
8.5 Shapes of s Orbitals

- Orbital shapes are most directly affected by the value of ℓ .
- s orbitals ($\ell = 0$)** are spherically symmetric and lack directionality; the probability of finding an electron is uniform on each sphere around the nucleus.

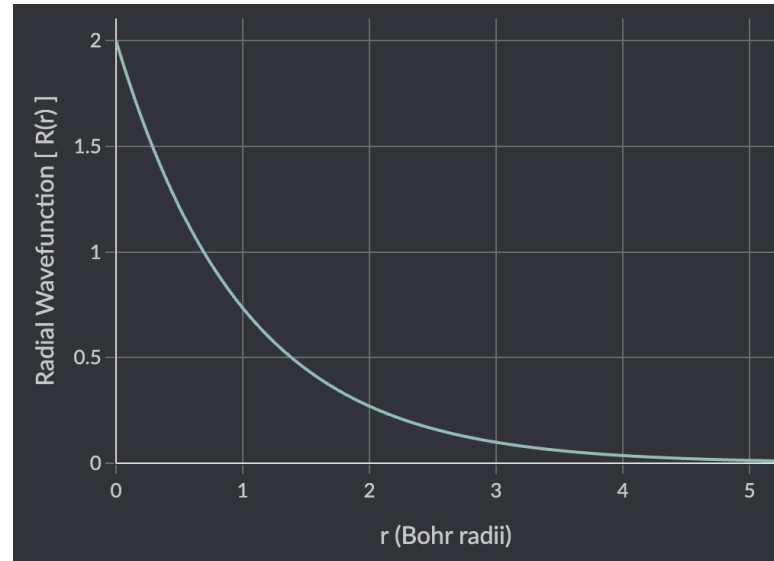


Try it yourself!

1s Orbital



Isosurface



Radial Distribution

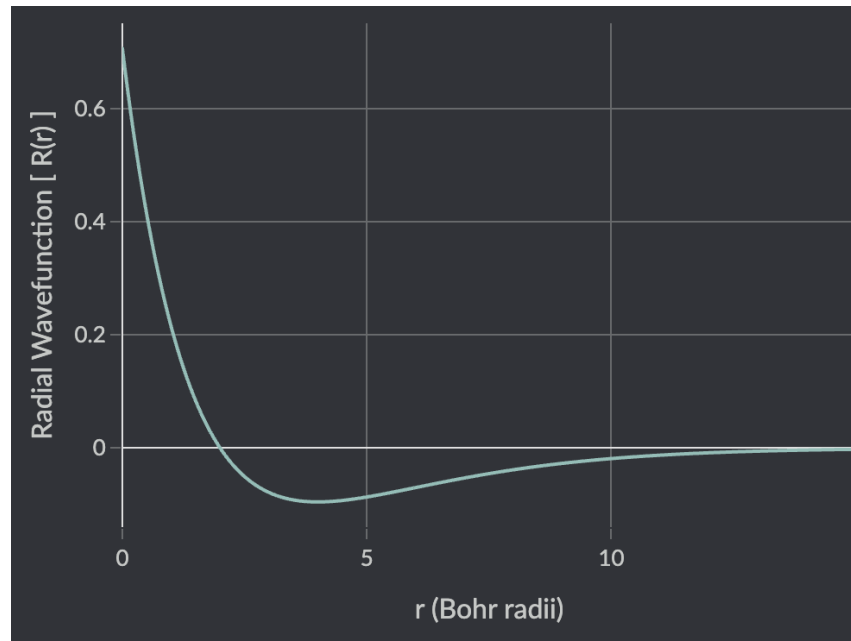
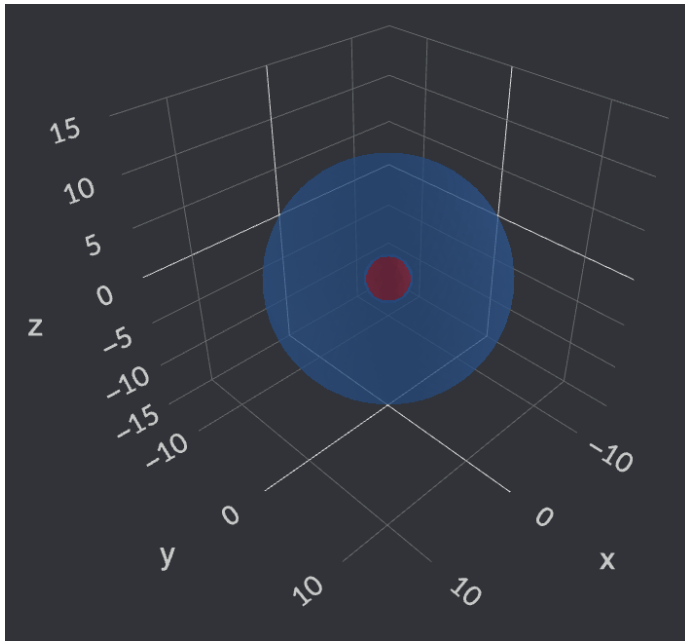
8.5 The 2s Orbital; Nodes

- The 2s orbital includes a sphere where the wave function changes sign from positive to negative; these surfaces where $\psi = 0$ are called **nodes**.
- Spherically shaped nodes are called **radial nodes** because they correspond to a single radius.



Try it yourself!

2s Orbital



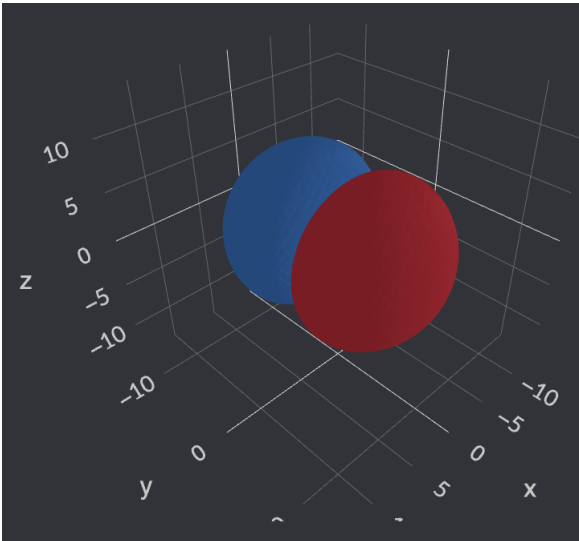
8.5 Shapes of p Orbitals

- **p orbitals** ($\ell = 1$) are dumbbell-shaped; there are three per subshell ($m_\ell = -1, 0, +1$).
- p orbitals include one nodal plane between the two **lobes** of the dumbbell.
- Planar and cone-shaped nodes are called **angular nodes** because they correspond to a single solid angle.

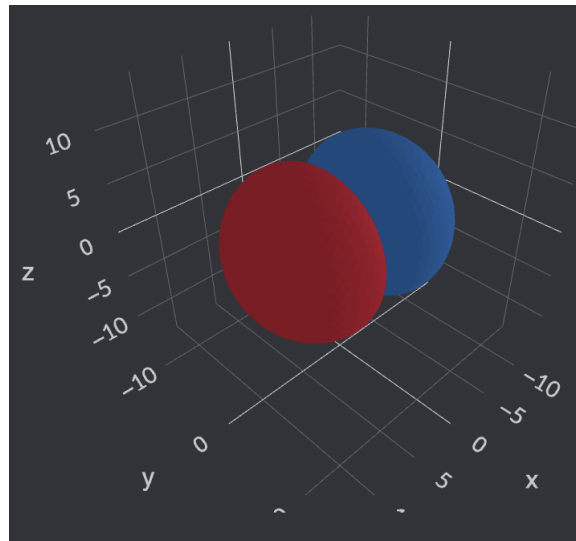


Try it yourself!

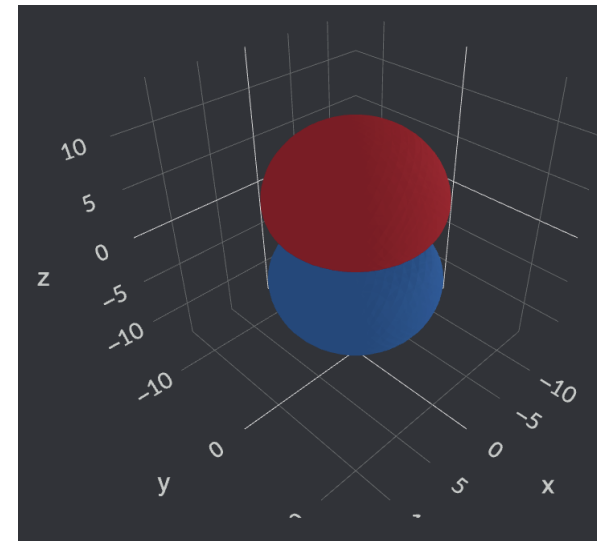
$2p_x$ Orbital



$2p_y$ Orbital

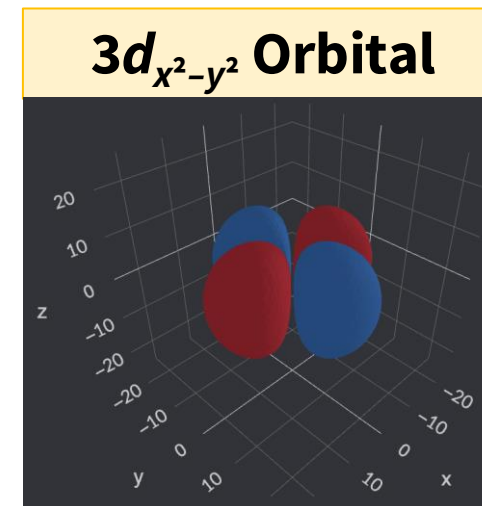
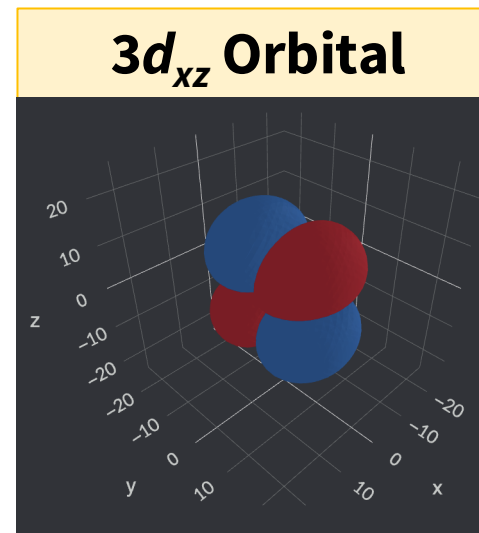
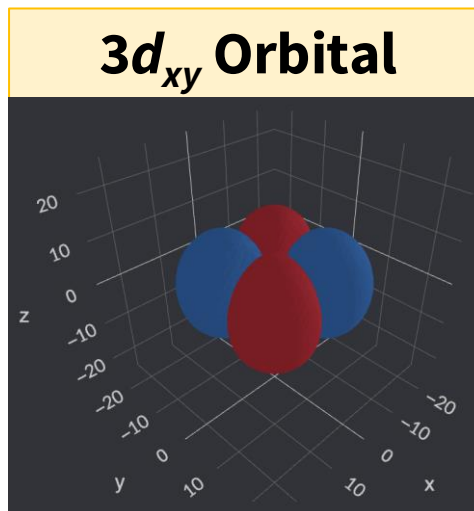


$2p_z$ Orbital

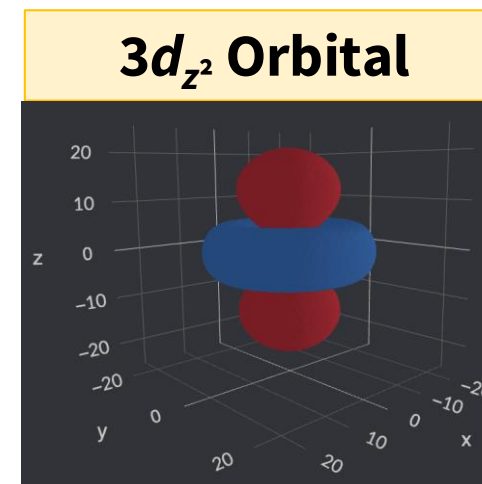
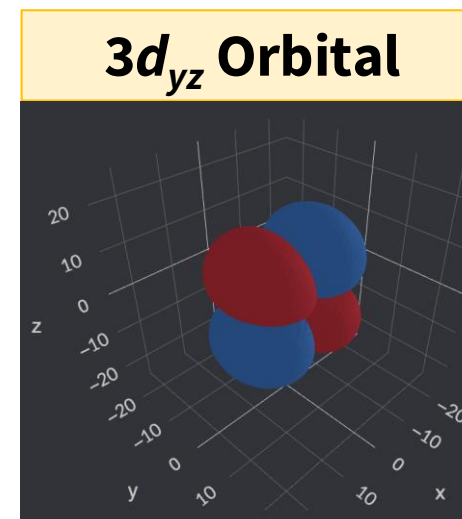


8.5 Shapes of d Orbitals

- d orbitals ($\ell = 2$)** are mostly cloverleaf-shaped; there are five per subshell ($m_\ell = -2, -1, 0, +1, +2$).



- d orbitals include two angular nodes (are we sensing a pattern here...?)



8.5 Shapes of f Orbitals

- **f orbitals ($\ell = 3$)** are visually complicated; there are seven per subshell ($m_\ell = -3, -2, -1, 0, +1, +2, +3$).
- f orbitals contain three angular nodes ($\ell = 3$).

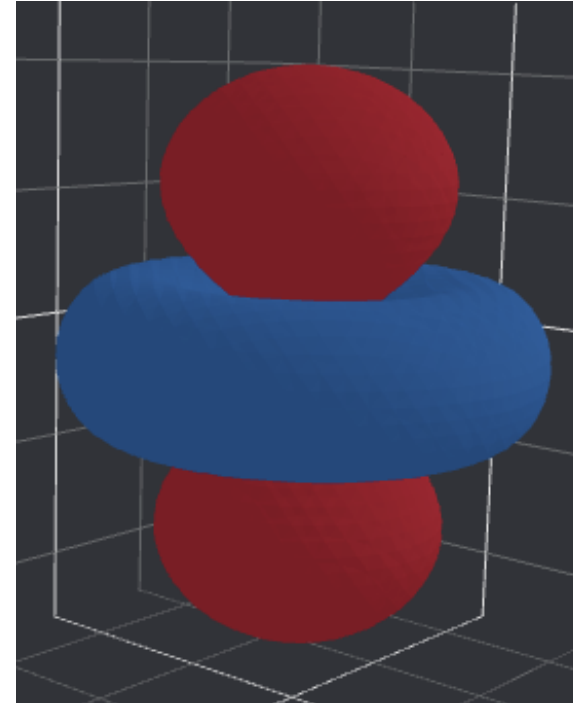


Try it yourself!

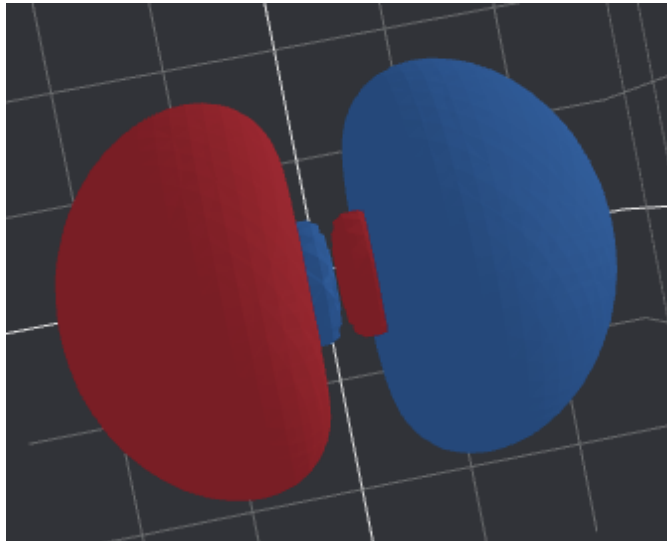
- Notes on the relative sizes and energies of orbitals...
 - As the value of n increases, orbital size and energy generally increase
 - Within a shell, orbital energy and size increase with ℓ : $s \rightarrow p \rightarrow d \rightarrow f$

Propose a set of quantum numbers that could be associated with the orbital below.

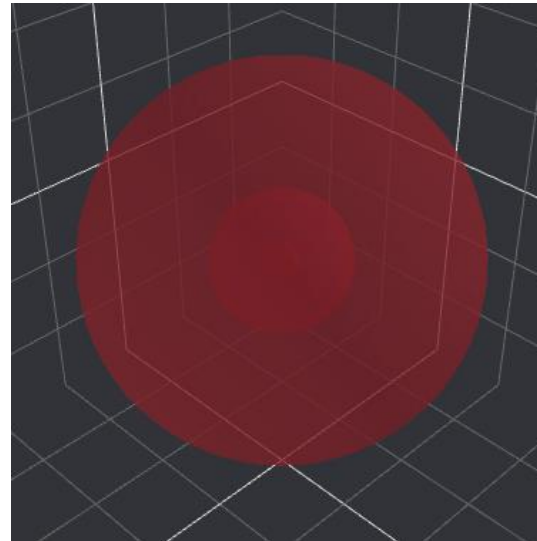
- A. $n = 2, l = 2, m_l = -1, m_s = +1/2$
- B. $n = 4, l = 3, m_l = +2, m_s = -1/2$
- C. $n = 4, l = 2, m_l = +3, m_s = +1/2$
- D. $n = 3, l = 2, m_l = 0, m_s = -1/2$



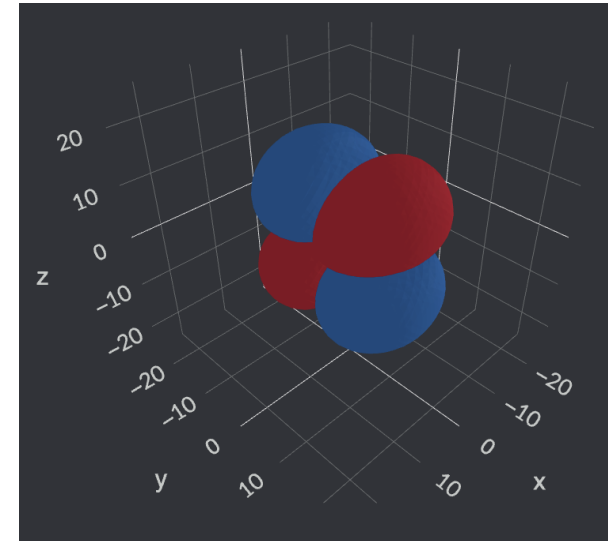
Arrange the orbitals in order of increasing energy within the $n = 3$ shell. Enter your response as a string of letters from lowest to highest energy, such as “ABC.”



Orbital A



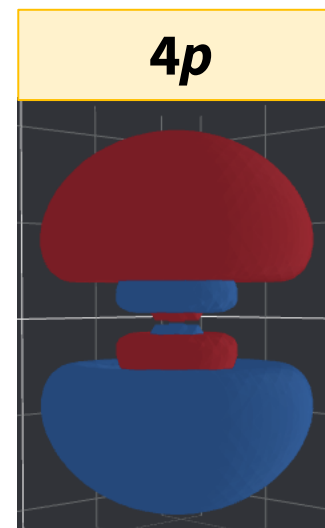
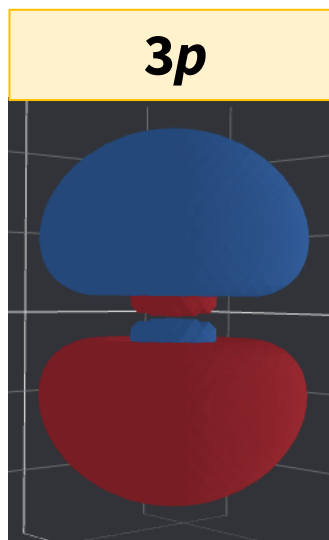
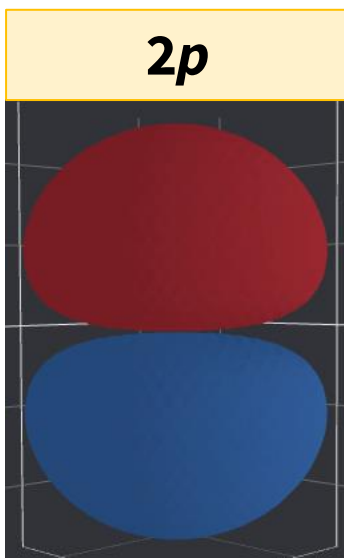
Orbital B



Orbital C

8.5 Using Quantum Numbers to Count Nodes

- A few important notes on n , ℓ , and nodes...
 - The number of angular nodes in an orbital is equal to ℓ .
This is one reason it's called the *angular* momentum quantum number.
 - The *total* number of nodes in an orbital is equal to $n - 1$.
 - Thus, the number of radial nodes in an orbital is equal to $n - \ell - 1$.
- **Example.** How many angular and radial nodes are in the orbitals below?



Which statement(s) is/are *true* regarding nodes? Select all that apply.

- A. All orbitals in the $n = 4$ energy level have three radial nodes.
- B. A 4s orbital has three radial nodes.
- C. The number of radial nodes depends only on the energy level of the orbital.
- D. The number of radial nodes depends on both the energy level and the subshell.
- E. There is zero probability of finding an electron in a radial node.
- F. Radial nodes are characteristic of the shape of an orbital.

Chapter 8.E

The Quantum Model of the Atom

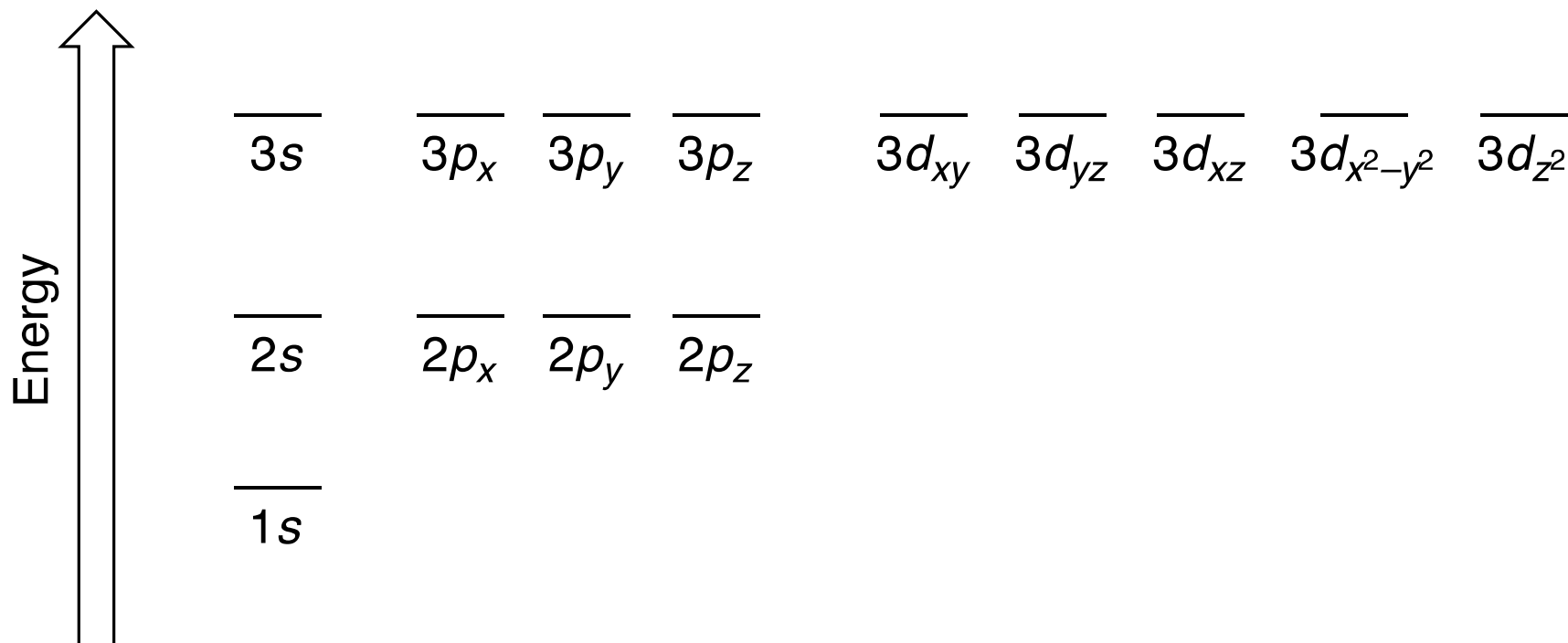
Chemical Principles I

CHEM 1211K

1. Describe and apply the dual nature of light.
2. Calculate energy, frequency, or wavelength of a sample of light.
3. Use the Bohr model of energy levels in atoms to explain light emission and absorption by gaseous atoms.
4. Calculate the energy or wavelength of a specific electronic transition in a hydrogen atom.
5. Use the concept of quantum numbers to characterize the energies, spatial distributions, and spins of atomic electrons.
6. Visualize and describe the spatial distributions and energies of atomic electrons using the quantum model of the atom.
7. Sketch the electronic energy levels of an atom.
8. Add electrons to atomic orbital energy (energy level) diagrams.
9. Determine electron configurations for the elements using the concepts of shells, subshells, and orbitals.

8.6 Orbital Energy Diagrams

- The occupancies and energies of atomic orbitals are represented on an **atomic orbital energy diagram**.
 - Energy is the vertical axis.
 - Orbitals are represented with horizontal lines and labels.
 - Electrons are represented with arrows pointing up ($m_s = +\frac{1}{2}$) or down ($m_s = -\frac{1}{2}$).
- Example.** For a neutral hydrogen atom, orbital energy depends only on n .



8.6 Atomic Orbitals for Heavy Atoms

- For heavier atoms, orbital energy depends on n and ℓ . The diagram below is a “standard scaffold” for atomic orbitals of these elements.
- We’ll add electrons to this scaffold using two ideas...
 - Aufbau principle:**

- Hund’s rule:**

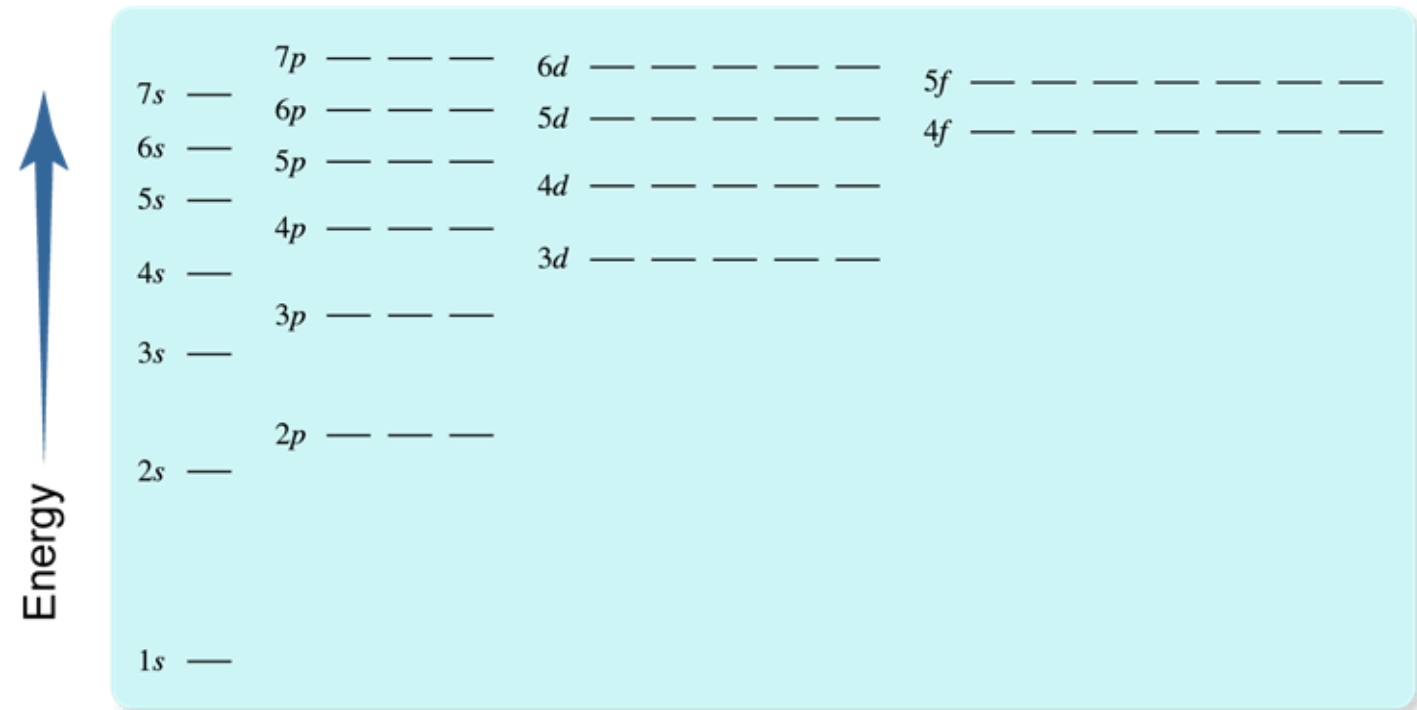


Figure 8.16. Typical energetic ordering of subshells for neutral atoms heavier than hydrogen.

Consider a d subshell containing 7 electrons. How many *unpaired* electrons does this subshell contain?

- The **electron configuration** of an atom concisely lists the orbitals that electrons occupy in that atom.
- We are often (but not always) interested in the **ground-state** electron configuration, which is lowest in energy. This configuration reflects the Aufbau principle and Hund's rule.
- **Example.** What is the ground-state electron configuration of a neutral sulfur atom?

8.7 The Atomic Orbital “Scaffold” and the Periodic Table

- Orbitals within a subshell are at the same energy (**degenerate**).
- The structure of the periodic table reflects the energetic ordering below...!

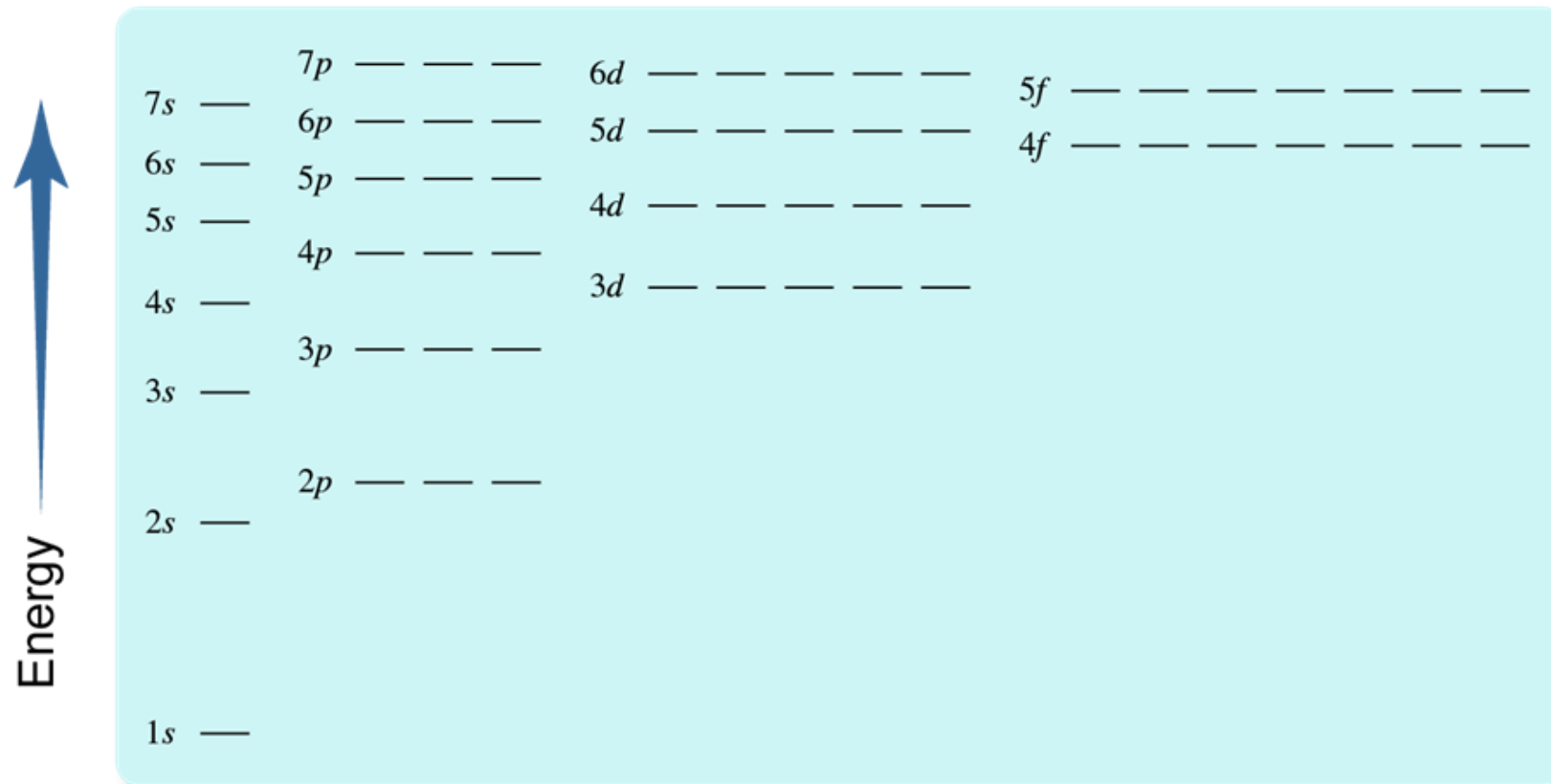


Figure 8.16. Typical energetic ordering of subshells for neutral atoms heavier than hydrogen.

8.7 Blocks on the Periodic Table

- The s , p , d , and f blocks are named after the highest-energy occupied subshell in atoms of the elements in these blocks.
- For s - and p -block elements, n of the highest-energy occupied subshell is equal to the period number.
- For d -block elements, n of the highest-energy occupied subshell is equal to the period number minus 1.
- For d -block elements, n of the highest-energy occupied subshell is equal to the period number minus 2.

	1	2														18
1	1s															1s
2	2s															
3	3s	3	4	5	6	7	8	9	10	11	12					
4	4s						3d									
5	5s						4d									
6	6s	*					5d									
7	7s	†					6d									
6	*							4f								
7	†							5f								

Figure 8.25. The highest-energy occupied subshell for each element on the periodic table.

8.7 Practice with Electron Configurations

Sapling Learning

Level 1 | Level 2 | Level 3

Orbitals

3p_z
3p_y
3p_x
3s
2p_z
2p_y
2p_x
2s
1s

Assemble a Beryllium Atom

4
Be

Orbital Diagram

Energy ↑

3p 3s 3p 3p_x 3p_y 3p_z
2p 2s 2p 2p_x 2p_y 2p_z
1s 1s

Electrons

•

Lewis Valence Electron Dot Structure

Be

Electron Configuration

Drag an orbital onto the stage. Then drag the correct number of electrons into the orbital. Continue the process until you have assembled the given atom. Make sure to select and fill orbitals in the correct order. Click **Reset** to clear your work for this level.

☒ Help

Reset

Figure 8.27. Electron configuration interactive. Build your own atoms!

Write the *full* ground-state electron configuration for antimony (Sb), element 51. How many electrons in antimony occupy s orbitals? Consider all occupied shells!

8.7 Abbreviated Electron Configurations



- **Abbreviated electron configurations:**

- **Example.** Write the abbreviated ground-state electron configurations for calcium and scandium atoms.

8.7 Electron Configurations of Cations



- **Cations** are lacking electrons relative to the neutral atom. The highest-energy electrons are “missing.”
- Main-group (*s*- and *p*-block) elements in groups 1, 2, and 13 form cations with the configuration of the closest noble gas.
- Transition (*d*-block) elements form cations that contain *only d* electrons; the *ns* orbital becomes higher in energy than the $(n - 1)d$ orbitals in transition metal cations.

8.7 Electron Configurations of Anions



- **Anions** have an excess of electrons relative to the neutral atom. The valence shell contains “extra” electrons.
- Main-group (*s*- and *p*-block) elements in groups 14 – 17 that are nonmetals form anions with the configuration of the closest noble gas.

How many *unpaired* electrons are in the ground state electron configuration for Fe^{3+} ?

An atom or ion has the abbreviated electron configuration [Kr]. Select the species that it could *NOT* possibly be.



- The highest-energy occupied shell is called the **valence shell**; electrons within are called **valence electrons**.
 - Always written explicitly in electron configurations
 - Outermost electrons involved in bonding and electron-transfer processes

- Lower-energy shells closer to the nucleus are called **core shells**; electrons within are called **core electrons**.
 - “Packed” into noble gas symbols when writing abbreviated configurations
 - Essentially just follow the nucleus everywhere without bonding or reacting

- Some of the amazing things we can now do with the quantum model of the atom...
 1. Relate frequency, wavelength, speed, and energy for light.
 2. Visualize diffraction, reflection, absorption, and emission using a classical wave model of light.
 3. Describe light using the particulate representation of photons; apply the particle model of light to describe absorption and emission by atoms and molecules.
 4. Interpret, describe, and rationalize the absorption and emission spectra of gaseous atoms.
 5. Apply the Bohr model and the Rydberg equation to describe the absorption and emission of specific wavelengths of light by the hydrogen atom.
 6. Describe and apply constraints on quantum numbers, which give rise to a scaffold of atomic orbitals reflected in the structure of the periodic table.
 7. Visualize atomic orbitals and interpret the shapes of atomic orbitals.
 8. Write electron configurations for neutral atoms, cations, and anions in the main group, as well as transition metal cations.